Weekly Planner

Week #: 9 (26/10/2026 To 30/10/2026)

Subject: Pre-Calculus Math **Unit Title: Functions, Graphs, Transformations, and Inverses** **Grade: 11**

Lessons	Learning targets	Assessment tools and strategies Formative/ Summative/ Differentiation	Resources and links for Parents
Lesson 1 Transformations of Functions (Translations, Reflections, and Stretches)	- I can describe and apply translations, reflections, stretches, and compressions to parent functions and predict their effects on graphs.	2.1 Clarify vocabulary and symbols using visuals, connections, and supports. 2.5 Use multiple media to illustrate concepts (diagrams, videos, models). 4.2 Ensure access to tools and assistive technologies.	• Khan Academy: Parent Functions • Khan Academy: Transforming Functions • Khan Academy: Composite Functions • Khan Academy: Inverse Functions • Desmos Graphing Calculator • Desmos Activity Builder
Lesson 2 Combining Transformations	- I can apply multiple transformations to graph and analyze transformed functions.		
Lesson 3 Function Operations and Composition	- I can perform operations on functions, determine domains of combined functions, and evaluate composite functions.		
Lesson 4 Inverse Functions	- I can determine whether a function has an inverse, find inverse functions algebraically, and verify inverses graphically and algebraically.		
Homework	- Textbook Sections 1-5 to 1-7 - Complete assigned Desmos activities - Khan Academy practice on transformations, composition, and inverse functions		
Teacher's notes/ Reminders	• Check **Microsoft Teams regularly** for class posts, announcements, resources, and assignments. • Complete assigned **Khan Academy** practice and extension activities. • Bring a charged device for graphing and technology-based investigations. • Review vocabulary including function, relation, continuity, limit, extrema, and rate of change. • Practice explaining solutions using appropriate mathematical notation and justification.		

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Technology Integration	• **Desmos.com**, Microsoft Teams, Graphing Calculator, GeoGebra
Cross-curricular links	**Physics:** Motion graphs and interpreting changing quantities. **Art and Design:** Symmetry, reflections, and scaling transformations. **Business Studies:** Multi-step cost, revenue, and profit models. **Computer Science:** Encoding, decoding, and reversible processes.
External Assessment Integration	Students will engage with: • **SAT/ACT/AP Precalculus Preparation:** Students analyze function behavior, interpret graphs, identify continuity, determine rates of change, and solve higher-order reasoning problems involving multiple representations.
Wellbeing/ ATL Skills	• **Critical Thinking:** Analyzing relationships and function behavior. • **Communication Skills:** Justifying mathematical conclusions using evidence. • **Research Skills:** Using technology to investigate and verify conjectures. • **Self-Management:** Monitoring progress and reflecting on problem-solving strategies. • **Thinking Skills:** Connecting graphical, numerical, and algebraic representations.
Standards	• **CCSS.HSF-IF.B.4** Interpret key features of graphs and relate them to quantities. • **CCSS.HSF-IF.C.7** Graph functions and show key features graphically. • **CCSS.HSF-BF.B.3** Identify the effect of transformations on graphs. • **CCSS.HSF-BF.A.1** Write a function that describes a relationship between two quantities. • **CCSS.HSF-BF.B.1** Combine standard function types algebraically. • **CCSS.HSF-BF.B.4** Find inverse functions. • **CCSS.HSF-IF.C.7b** Graph inverse relationships.
Inquiry/ TOK Question	• To what extent do mathematical functions accurately represent and predict behavior in the real world?
Success Criteria (Measurable)	Lesson 1: Transformations of Functions ■ All students can identify and apply vertical and horizontal translations. ● Most students should apply reflections, stretches, and compressions. ▲ Some students could write transformed equations directly from graphs and vice versa. Lesson 2: Combining Transformations ■ All students can identify multiple transformations on a function. ● Most students should graph functions involving combinations of transformations. ▲ Some students could explain the sequence of transformations and predict resulting graphs without graphing technology.

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	Lesson 3: Function Operations and Composition ■ All students can add, subtract, multiply, and divide functions. ● Most students should determine domains of combined functions and evaluate composite functions. ▲ Some students could model real-world situations using composite functions. Lesson 4: Inverse Functions ■ All students can identify one-to-one functions and determine whether inverses exist. ● Most students should find inverse functions algebraically and graphically. ▲ Some students could verify inverses using composition and explain domain-range relationships.
National Identity Domain	Conservation الحفاظ علي البيئة Global Understanding التفاهم العالمي Heritage التراث
	Area: UAE Innovation and Future Readiness Area: UAE culture and heritage Area: UAE Economic Development **Area:** Responsible Citizenship and Innovation
	Implementation: • Students examine how mathematical models are used in UAE smart-city projects and data-driven planning initiatives. • Students investigate geometric patterns and symmetry in traditional Emirati architecture and Islamic design. • Students analyze simplified models related to tourism growth, transportation systems, and business productivity in the UAE. • Students explore how inverse processes are used in digital systems, cybersecurity, and technological innovation supporting the UAE's vision for a knowledge-based economy.